

Emergent Suspiciousness in Trained Monitors

Prompt Injection Shortcuts,
Context-Dependent Robustness, and Data Augmentation

Jeremias Ferrao Michael Kenny

4 October 2026

Our approach: distill the monitoring decision directly

We reuse Sinha et al.'s trajectory data framework: 8,688 training examples.

Sinha et al.: SFT + RL

SFT on monitor rationales and final 0–10 scores.

RL rewards calibration and ranking against binary labels.

Output: a rationale and score.

Mean OOD pAUROC@20: 0.684

Ours: logit distillation

Distill Kimi K3's soft harmfulness probabilities into a smaller monitor.

Train the binary decision logits, without rationale targets or RL.

Output: a harmfulness probability.

Mean OOD pAUROC@20: 0.776

One illustrative comparison: both use Qwen3.5-4B on six OOD sources.

Sinha et al. (2026), *Training Deliberative Monitors for Black-Box Scheming Detection*.